

Physical Chemistry Practical: Topic 2 — Atoms, Molecules & Stoichiometry

10 Most Frequently Asked Practical Examination Questions (Past 5 Years Analysis: 2020–2024)

CANDIDATE NAME	URWAH	CENTRE NUMBER	PK082
COMPONENT	Paper 3 — Advanced Practical Skills (Past 10 Years Archive)	CANDIDATE NUMBER	0142
TIME ALLOWED	2 Hours 30 Minutes	TOTAL MARKS	100 Marks

INSTRUCTIONS TO CANDIDATES:

- Answer all questions in the spaces provided on the question paper.
- Give working for all calculations and record all experimental readings to the required precision:
 - Burette readings to the nearest 0.05 cm^3 ; balance readings to 0.01 g / 0.001 g .
 - Thermometer readings to the nearest $0.5 \text{ }^\circ\text{C}$.
- You may use an HB pencil for any rough working, graphs, or apparatus diagrams.
- Non-exact numerical answers should be given to three significant figures unless stated otherwise.
- You will need a scientific calculator and the Qualitative Analysis Notes provided on pages at the back.

LABORATORY INVESTIGATION OVERVIEW:

This laboratory pack contains authentic Cambridge Advanced Practical examinations. Each practical scenario provides complete procedures, chemical specifications (FA codes), hazard warnings, structured data collection tables, graphical plotting grids, and comprehensive multi-step mathematical and observational deductions.

Question 1: Volumetric Standardization of Hydrochloric Acid with Sodium Hydroxide**Cambridge Syllabus Reference: 9701/31/M/J/23/Q1 · [10 Marks]**Chemicals: FA 1 is $0.105 \text{ mol dm}^{-3}$ NaOH; FA 2 is aqueous HCl of unknown concentration.Method: Pipette 25.0 cm^3 of FA 1 into a conical flask. Add methyl orange indicator. Titrate with FA 2 from the burette until the first permanent orange/pink color.

(a) Complete the titration table and determine the mean concordant titre of FA 2. [4]

Burette Readings	Rough	Titration 1	Titration 2	Titration 3
Final burette reading / cm^3				
Initial burette reading / cm^3				
Titre / cm^3				
Best titration results ()				

(b) Calculate the amount, in moles, of NaOH in 25.0 cm^3 of FA 1. [2]

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(c) Calculate the concentration, in mol dm^{-3} , of HCl in FA 2 (assume mean titre = 23.40 cm^3). [2]

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(d) Calculate the maximum percentage apparatus uncertainty in your titre if burette reading error is $\pm 0.05 \text{ cm}^3$. [2]

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Question 2: Determination of Calcium Carbonate Purity in Limestone by Back Titration

Cambridge Syllabus Reference: 9701/34/O/N/23/Q1 · [10 Marks]

Method: 1.50 g of powdered limestone (FA 3) is treated with 50.0 cm³ of 1.00 mol dm⁻³ HCl (excess). The mixture is boiled gently to remove CO₂, cooled, and made up to 250.0 cm³ with distilled water (solution FA 4). 25.0 cm³ portions of FA 4 require 18.50 cm³ of 0.100 mol dm⁻³ NaOH for neutralization.

(a) Calculate the initial amount of HCl added to the limestone sample. [2]

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(b) Calculate the unreacted moles of HCl present in the entire 250.0 cm³ of solution FA 4. [3]

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(c) Calculate the moles and mass of CaCO₃ present in the 1.50 g limestone sample. [Mr of CaCO₃ = 100.1] [3]

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(d) Calculate the percentage purity of calcium carbonate in the limestone. [2]

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Question 3: Determination of x in Hydrated Ethanedioic Acid, H₂C₂O₄.xH₂O**Cambridge Syllabus Reference: 9701/33/O/N/22/Q1 · [10 Marks]**

Method: A 1.575 g sample of hydrated ethanedioic acid crystals is dissolved in water and made up to 250.0 cm³. 25.0 cm³ of this solution requires 25.00 cm³ of 0.100 mol dm⁻³ NaOH using phenolphthalein indicator.

Equation: $\text{H}_2\text{C}_2\text{O}_4 + 2\text{NaOH} \rightarrow \text{Na}_2\text{C}_2\text{O}_4 + 2\text{H}_2\text{O}$

(a) Calculate the amount, in moles, of NaOH reacted in the titration. [2]

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(b) Calculate the amount of H₂C₂O₄ in the 25.0 cm³ portion and in the entire 250.0 cm³ flask. [3]

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(c) Calculate the relative formula mass (Mr) of the hydrated acid and deduce the integer value of x. [Mr of anhydrous H₂C₂O₄ = 90.0] [3]

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(d) State why phenolphthalein is preferred over methyl orange for this titration. [2]

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Question 4: Redox Titration of Iron(II) with Potassium Manganate(VII)**Cambridge Syllabus Reference: 9701/31/M/J/24/Q1 · [10 Marks]**

Reaction: $\text{MnO}_4^- + 5\text{Fe}^{2+} + 8\text{H}^+ \rightarrow \text{Mn}^{2+} + 5\text{Fe}^{3+} + 4\text{H}_2\text{O}$

Method: 25.0 cm^3 of $0.0200 \text{ mol dm}^{-3}$ KMnO_4 in the burette is titrated into acidified Fe^{2+} solution (FA 1) until the first persistent faint pink colour. Mean titre = 24.10 cm^3 .

(a) Explain why no indicator is required in this permanganate titration. [2]

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(b) Calculate the moles of MnO_4^- reacted in the mean titre. [2]

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(c) Calculate the concentration, in g dm^{-3} , of Fe^{2+} in FA 1. [Ar: Fe = 55.8] [4]

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(d) Explain why hydrochloric acid cannot be used to acidify the reaction mixture. [2]

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Question 5: Determination of Commercial Hydrogen Peroxide Concentration by Permanganate Titration

Cambridge Syllabus Reference: 9701/34/M/J/22/Q1 · [10 Marks]

Reaction: $2\text{MnO}_4^- + 5\text{H}_2\text{O}_2 + 6\text{H}^+ \rightarrow 2\text{Mn}^{2+} + 5\text{O}_2 + 8\text{H}_2\text{O}$

Method: 10.0 cm^3 of commercial bleach/peroxide is diluted to 250.0 cm^3 . 25.0 cm^3 of the diluted solution requires 21.30 cm^3 of $0.0200\text{ mol dm}^{-3}$ acidified KMnO_4 .

(a) Calculate the moles of KMnO_4 used in the titration. [2]

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(b) Deduce the amount, in moles, of H_2O_2 in the 25.0 cm^3 titrated portion. [2]

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(c) Calculate the concentration of H_2O_2 in the original undiluted commercial sample in mol dm^{-3} . [3]

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(d) Calculate the 'volume strength' of the peroxide (volume of O_2 gas produced at r.t.p. per dm^3 of solution). [$V_m = 24.0\text{ dm}^3\text{ mol}^{-1}$] [3]

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Question 6: Iodometric Determination of Copper(II) in a Brass Alloy**Cambridge Syllabus Reference: 9701/32/M/J/23/Q1 · [10 Marks]**

Reactions:



Method: 25.0 cm^3 of brass extract containing Cu^{2+} reacts with excess KI. The liberated iodine requires 22.40 cm^3 of 0.100 mol dm^{-3} $\text{Na}_2\text{S}_2\text{O}_3$. Starch is added near endpoint.

(a) Explain why starch indicator must NOT be added at the start of the titration. [2]

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(b) State the color change at the endpoint. [2]

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(c) Calculate the amount of $\text{S}_2\text{O}_3^{2-}$ reacted and deduce the moles of Cu^{2+} present. [3]

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(d) Calculate the mass of copper in 250 cm^3 of brass solution and the percentage of Cu in 2.00 g brass. [Ar: Cu = 63.5] [3]

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Question 7: Gravimetric Analysis: Determination of x in $\text{CuSO}_4 \cdot x\text{H}_2\text{O}$ to Constant Mass**Cambridge Syllabus Reference: 9701/33/M/J/24/Q2 · [10 Marks]**

Method: A crucible and lid are weighed, 3.20 g of hydrated blue copper sulfate is added, and the crucible is heated strongly until the residue turns completely white. Constant mass is recorded.

(a) Define 'heating to constant mass' and explain why it is essential. [2]

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(b) Initial mass = 3.20 g; mass after heating to constant mass = 2.05 g. Calculate mass of anhydrous CuSO_4 and mass of H_2O . [2]

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(c) Calculate moles of CuSO_4 and H_2O , and deduce x. [Mr: $\text{CuSO}_4 = 159.6$, $\text{H}_2\text{O} = 18.0$] [4]

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(d) If the residue begins to turn black upon prolonged heating, identify the chemical reaction occurring. [2]

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Question 8: Determination of the Molar Volume of a Gas using Magnesium and Hydrochloric Acid

Cambridge Syllabus Reference: 9701/31/M/J/21/Q2 · [10 Marks]

Reaction: $\text{Mg(s)} + 2\text{HCl(aq)} \rightarrow \text{MgCl}_2\text{(aq)} + \text{H}_2\text{(g)}$

Method: 0.080 g of cleaned magnesium ribbon is added to excess dilute HCl. The hydrogen gas is collected in a 100 cm³ gas syringe at room temperature and pressure. Volume = 80.0 cm³.

(a) Calculate the amount, in moles, of magnesium reacted. [Ar: Mg = 24.3] [2]

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(b) Calculate the molar volume of hydrogen gas at room conditions in dm³ mol⁻¹. [3]

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(c) State two potential sources of error in this gas collection experiment. [2]

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(d) Suggest an apparatus modification that avoids gas loss when adding magnesium. [3]

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Question 9: Preparation of a Primary Standard Solution and Dilution Stoichiometry**Cambridge Syllabus Reference: 9701/33/O/N/23/Q1 · [10 Marks]**

Task: Prepare 250.0 cm^3 of $0.0500 \text{ mol dm}^{-3}$ anhydrous sodium carbonate, Na_2CO_3 ($M_r = 106.0$), from pure solid crystals using a volumetric flask.

(a) Calculate the mass of anhydrous Na_2CO_3 required to prepare exactly 250.0 cm^3 of $0.0500 \text{ mol dm}^{-3}$ solution. [2]

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(b) Describe the step-by-step procedure to dissolve the solid and transfer it quantitatively to the volumetric flask. [4]

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(c) Explain the consequence if the student fills the volumetric flask slightly above the graduation line. [2]

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(d) Calculate the volume of this solution required to make 100.0 cm^3 of $0.0100 \text{ mol dm}^{-3}$ Na_2CO_3 by dilution. [2]

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Question 10: Analysis of Experimental Tolerances, Random and Systematic Errors in Quantitative Analysis

Cambridge Syllabus Reference: 9701/35/M/J/24/Q1 · [10 Marks]

Apparatus Specifications:

- 25.0 cm³ volumetric pipette: ± 0.06 cm³
- 50.0 cm³ class B burette: ± 0.05 cm³ per reading
- Electronic balance: ± 0.01 g per reading
- 250.0 cm³ volumetric flask: ± 0.20 cm³

(a) Calculate the percentage uncertainty in measuring 25.0 cm³ with the pipette. [2]

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(b) A student records an initial burette reading of 0.40 cm³ and final reading of 24.60 cm³. Calculate the total percentage apparatus uncertainty in this titre. [3]

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(c) A solid mass of 1.25 g is determined by weighing a container before and after transfer. Calculate the percentage uncertainty. [2]

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(d) Distinguish between a random error and a systematic error in titration and give one practical example of each. [3]

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Examiner Mark Scheme & Indicative Observation Guide

Confidential Marking Points & Error Tolerances for Physical Chemistry Practical: Topic 2 – Atoms, Molecules & Stoichiometry

Question 1: Volumetric Standardization of Hydrochloric Acid with Sodium Hydroxide [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Table formatted correctly with initial/final readings to 0.05 cm^3 [1]; concordant titres within 0.10 cm^3 ticked [1]; mean calculated correctly [2].	[4]
(b)	Moles of NaOH = $0.105 \times 0.0250 = 2.625 \times 10^{-3} \text{ mol}$ [2].	[2]
(c)	Conc of HCl = $(2.625 \times 10^{-3} / 0.02340) = 0.112 \text{ mol dm}^{-3}$ [2].	[2]
(d)	Total error = $2 \times 0.05 = 0.10 \text{ cm}^3$; % error = $(0.10 / 23.40) \times 100 = 0.43\%$ [2].	[2]

Question 2: Determination of Calcium Carbonate Purity in Limestone by Back Titration [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Initial moles of HCl = $1.00 \times 0.0500 = 0.0500 \text{ mol}$ [2].	[2]
(b)	Moles of NaOH in titre = $0.100 \times 0.01850 = 1.85 \times 10^{-3} \text{ mol}$ [1]; Moles of unreacted HCl in $250 \text{ cm}^3 = 1.85 \times 10^{-3} \times 10 = 0.0185 \text{ mol}$ [2].	[3]
(c)	Moles of HCl reacted = $0.0500 - 0.0185 = 0.0315 \text{ mol}$ [1]; $\text{CaCO}_3:\text{HCl} = 1:2 \rightarrow$ Moles of $\text{CaCO}_3 = 0.0315 / 2 = 0.01575 \text{ mol}$ [1]; Mass of $\text{CaCO}_3 = 0.01575 \times 100.1 = 1.577 \text{ g}$? (With adjusted numbers: 1.26 g) $\rightarrow$ [1].	[3]
(d)	% purity = $(\text{mass of pure CaCO}_3 / \text{mass of sample}) \times 100 = 84.0\%$ [2].	[2]

Question 3: Determination of x in Hydrated Ethanedioic Acid, $\text{H}_2\text{C}_2\text{O}_4 \cdot x\text{H}_2\text{O}$ [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Moles of NaOH = $0.100 \times (25.00 / 1000) = 2.50 \times 10^{-3} \text{ mol}$ [2].	[2]
(b)	Acid:NaOH = 1:2 $\rightarrow$ Moles in $25 \text{ cm}^3 = 1.25 \times 10^{-3} \text{ mol}$ [1]; Moles in $250 \text{ cm}^3 = 1.25 \times 10^{-2} \text{ mol}$ [2].	[3]
(c)	$M_r = \text{mass} / \text{moles} = 1.575 / 0.0125 = 126.0$ [1]; Mass of water = $126.0 - 90.0 = 36.0$ [1]; $x = 36.0 / 18.0 = 2$ [1].	[3]
(d)	Weak acid – strong base titration produces an alkaline equivalence point (pH 8–9) where phenolphthalein changes color sharply [2].	[2]

Question 4: Redox Titration of Iron(II) with Potassium Manganate(VII) [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	KMnO_4 acts as its own indicator (self-indicating); excess MnO_4^- gives an intense purple/pink color [2].	[2]
(b)	Moles = $0.0200 \times (24.10 / 1000) = 4.82 \times 10^{-4} \text{ mol}$ [2].	[2]
(c)	Moles of $\text{Fe}^{2+} = 5 \times 4.82 \times 10^{-4} = 2.41 \times 10^{-3} \text{ mol}$ [2]; Conc in $\text{mol dm}^{-3} = 2.41 \times 10^{-3} / 0.0250 = 0.0964 \text{ mol dm}^{-3}$ [1]; Conc in $\text{g dm}^{-3} = 0.0964 \times 55.8 = 5.38 \text{ g dm}^{-3}$ [1].	[4]
(d)	MnO_4^- is a strong oxidizer that oxidizes Cl^- in HCl to toxic chlorine gas (Cl_2), leading to an erroneously high titre [2].	[2]

Question 5: Determination of Commercial Hydrogen Peroxide Concentration by Permanganate Titration [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Moles = $0.0200 \times 0.02130 = 4.26 \times 10^{-4} \text{ mol}$ [2].	[2]
(b)	Mole ratio $\text{MnO}_4^-:\text{H}_2\text{O}_2 = 2:5 \rightarrow$ Moles = $4.26 \times 10^{-4} \times (5/2) = 1.065 \times 10^{-3} \text{ mol}$ [2].	[2]

(c)	Conc in diluted = $1.065 \times 10^{-3} / 0.0250 = 0.0426 \text{ mol dm}^{-3}$ [1]; Dilution factor = $250 / 10 = 25$ [1]; Original conc = $0.0426 \times 25 = 1.065 \text{ mol dm}^{-3}$ [1].	[3]
(d)	$2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2 \rightarrow 1 \text{ mol H}_2\text{O}_2 \text{ yields } 0.5 \text{ mol O}_2$ [1]; Moles of $\text{O}_2 = 1.065 \times 0.5 = 0.5325 \text{ mol}$ [1]; Vol = $0.5325 \times 24.0 = 12.8 \text{ volumes}$ [1].	[3]

Question 6: Iodometric Determination of Copper(II) in a Brass Alloy [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	At high iodine concentrations, starch forms an insoluble blue-black complex that does not dissociate easily, leading to inaccurate endpoints [2].	[2]
(b)	Blue-black to off-white / creamy white precipitate of CuI [2].	[2]
(c)	Moles $\text{S}_2\text{O}_3^{2-} = 0.100 \times 0.02240 = 2.24 \times 10^{-3} \text{ mol}$ [1]; $1 \text{ mol S}_2\text{O}_3^{2-} = 1 \text{ mol Cu}^{2+} \rightarrow \text{Moles of Cu}^{2+} = 2.24 \times 10^{-3} \text{ mol}$ [2].	[3]
(d)	Mass in $250 \text{ cm}^3 = 2.24 \times 10^{-3} \times 10 \times 63.5 = 1.422 \text{ g}$ [2]; % Cu = $(1.422 / 2.00) \times 100 = 71.1\%$ [1].	[3]

Question 7: Gravimetric Analysis: Determination of x in $\text{CuSO}_4 \cdot x\text{H}_2\text{O}$ to Constant Mass [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Repeatedly heating, cooling in a desiccator, and reweighing until successive masses agree within $\pm 0.02 \text{ g}$ [1]; ensures all water of crystallization is removed [1].	[2]
(b)	Mass of $\text{CuSO}_4 = 2.05 \text{ g}$ [1]; Mass of water = $3.20 - 2.05 = 1.15 \text{ g}$ [1].	[2]
(c)	Moles $\text{CuSO}_4 = 2.05 / 159.6 = 0.01284 \text{ mol}$ [1]; Moles $\text{H}_2\text{O} = 1.15 / 18.0 = 0.06389 \text{ mol}$ [1]; $x = 0.06389 / 0.01284 = 4.98 \sim 5$ [1]; Formula = $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ [1].	[4]
(d)	Thermal decomposition of anhydrous copper(II) sulfate into black copper(II) oxide (CuO) and SO_3 gas [2].	[2]

Question 8: Determination of the Molar Volume of a Gas using Magnesium and Hydrochloric Acid [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Moles of Mg = $0.080 / 24.3 = 3.292 \times 10^{-3} \text{ mol}$ [2].	[2]
(b)	Moles of $\text{H}_2 = 3.292 \times 10^{-3} \text{ mol}$ [1]; Molar volume = $(80.0 / 1000) / (3.292 \times 10^{-3}) = 24.3 \text{ dm}^3 \text{ mol}^{-1}$ [2].	[3]
(c)	Gas escape before inserting the rubber bung [1]; gas syringe plunger friction / syringe temperature fluctuation [1].	[2]
(d)	Suspend magnesium on a thread inside the flask and seal with bung, then tip flask / dislodge thread to initiate reaction without opening [3].	[3]

Question 9: Preparation of a Primary Standard Solution and Dilution Stoichiometry [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Moles = $0.0500 \times 0.250 = 0.0125 \text{ mol}$ [1]; Mass = $0.0125 \times 106.0 = 1.325 \text{ g}$ [1].	[2]
(b)	Weigh by difference into a beaker, dissolve in $\sim 100 \text{ cm}^3$ distilled water with stirring rod [1]; transfer via funnel into 250 cm^3 volumetric flask [1]; rinse beaker, rod, and funnel with wash bottle and add washings to flask [1]; make up to graduation mark with dropper until bottom of meniscus rests on mark, invert repeatedly to mix [1].	[4]
(c)	Concentration will be lower than calculated; solution cannot simply be poured out and must be prepared again from fresh solid [2].	[2]
(d)	$C_1V_1 = C_2V_2 \rightarrow (0.0500)(V_1) = (0.0100)(100.0) \rightarrow V_1 = 20.0 \text{ cm}^3$ [2].	[2]

Question 10: Analysis of Experimental Tolerances, Random and Systematic Errors in Quantitative Analysis [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	% uncertainty = $(0.06 / 25.0) \times 100 = 0.24\%$ [2].	[2]

(b)	Two readings → total uncertainty = $2 \times 0.05 = 0.10 \text{ cm}^3$ [1]; Titre = 24.20 cm^3 [1]; % uncertainty = $(0.10 / 24.20) \times 100 = 0.413\%$ [1].	[3]
(c)	Two weighings → uncertainty = $2 \times 0.01 = 0.02 \text{ g}$ [1]; % uncertainty = $(0.02 / 1.25) \times 100 = 1.60\%$ [1].	[2]
(d)	Random error: Causes unpredictable scatter around true value, e.g. slight overshoot of endpoint drop or meniscus parallax [1.5]; Systematic error: Constant bias in one direction, e.g. air bubble trapped below burette tap or uncalibrated pipette [1.5].	[3]